

# **Wearable Robotic Trainer with AI-based Adaptive Motivation for Pediatric Gait Rehabilitation**



*A BS Final Year Project by*

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## Certificate of Approval

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**In the name of Allah (SWT), the most beneficent and the most merciful**

A BS Final Year Project submitted to the  
Department of Electrical and Computer Engineering  
International Islamic University, Islamabad  
In partial fulfillment of the requirements  
For the award of the degree of  
Bachelor of Science in Electrical Engineering

# **Declaration**

We hereby declare that this work, neither as a whole nor as a part thereof has been copied out from any source. No portion of the work presented in this report has been submitted in support of any application for any other degree or qualification of this or any other university or institute of learning. We further declare that the referred text is properly cited in the references.

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# Project Information

**Project Title:** Wearable Robotic Trainer with AI-based Adaptive Motivation for Pediatric Gait Rehabilitation

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**Tools Used:**

- ESP32 (Arduino IDE / C++)
- Node-RED (Therapist Dashboard)
- Node-red (AI Workflow Automation)
- Google Sheets / Apps Script
- Python

# **Abstract**

Motor deficit disabilities affecting children, such as Cerebral Palsy and developmental gait disorders, have an impact on many children globally and pose serious challenges to rehabilitation facilities. Conventional therapy treatments necessitate intensive one-on-one supervision by physical therapists in most cases, which makes it a very expensive and non-scalable treatment option. On top of that, a lot of the existing robotic rehabilitation systems are devoid of the adaptive engagement mechanisms which are behind the sustained participation of pediatric patients. Wearable robotic trainer with AI-based adaptive motivation for pediatric gait rehabilitation is the main focus of this thesis. The proposed system uses a lower-limb exoskeleton and physiological and biomechanical sensors for personalized rehabilitation support. A two-phase therapeutic protocol based on the Cuevas Medek Exercises (CME) framework is implemented, starting from Partial Body Weight Support (PBWS) and evolving to an assist-as-needed control strategy based on real-time muscle activity and load distribution measurements. To make the patient more involved and encourage, a motivational module that is driven by AI, continuously evaluates rehabilitation performance and gives personalized audio and visual feedback.

Moreover, the therapist is provided with a dashboard where he can remotely configure sessions, monitor in real-time, and record data. Experimental testing exhibited the reliable operation of the system, precise motion assistance, adequate user-intent detection, and positive interaction during rehabilitation sessions. The findings suggest that the integration of adaptive robotic assistance with intelligent motivational feedback can enhance both the accessibility and the effectiveness of pediatric gait rehabilitation, thus being a promising solution for the rehabilitation of future technologies.

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# **Chapter 1**

## **Introduction**

### **1.1 Background**

The key aim of pediatric gait rehabilitation is one of the major tasks of pediatric practice, namely to restore the ability to walk independently in children, who have experienced motor developmental problems because of different neurological or musculoskeletal disorders. Independent walking plays a significant role not only in physiological terms but also in psychological and social aspects. Although initial and continuous treatment is highly valuable from a clinical perspective, there exist some structural challenges to overcome in this domain – the lack of therapists, high cost of treatment and the absence of user-friendly online tools specifically designed for children.

Such an initiative called "Wearable Robotic Trainer with AI-based Adaptive Motivation for Pediatric Gait Rehabilitation" is the result of the director's personal experience working with high physical and emotional load put on the shoulders of physiotherapists in case of sessions with kids with motor and gait delays. The core concept underlying this work is that a wearable robot can be a helpful assistive device, which would decrease the physical load of treatment, perform clinically recommended training exercises and activate children through intelligent motivation.

#### **1.1.1 Motivation**

Inspiration for the current project was found in watching video footage of physical therapists treating children with motor disabilities through various social media networks. Noticing the amazing dedication of these professionals as well as the amazing perseverance of the children they treat was enough to find an interesting motivation to see how electrical and computer engineering can be used to assist the aforementioned process. It turns out that even though the dedication of the therapists is impressive, the whole process of gait rehabilitation, especially when conducted in a one-on-one setting, is mentally and physically exhausting and quite difficult to scale up. Another aspect is that children, especially the young ones, are challenging patients, as apart from being physically assisted, they also require constant emotional encouragement in order to stay active and engaged during all of the therapy sessions.

## 1.2 Project Overview

The project aims to develop a robotic rehabilitation system for pediatric gait training by integrating intelligent control, physiological sensing, and adaptive motivation in order to optimize therapeutic effectiveness and patient engagement.

Specifically, the project aims to:

- Objective: To design and fabricate a mechanically durable, hand-fabricated lower-limb exoskeleton for pediatric users (~ 3 years old) using mild-steel.
- Real-time EMG and load cell data feedback will allow a mode of operation based on two different therapeutic adaptations: (1) full support leading to an interactive gait training session and (2) Assist as needed, which that optimally scales the provided assistance in real time depending on the summed myogenic force.
- Synchronized control of multi joint servo motors for safe and stable gait execution, Design a real time embedded control system using ESP32.
- Integrate multimodal sensing, including EMG-based

### 1.2.1 System Architecture Overview

The system is organized into five functional layers. The Mechanical Layer consists of the manually fabricated mild-steel exoskeleton frame, hand-built by a local metalsmith, with bilateral hip, knee, and ankle joints sized for a three-year-old child. The Actuation Layer comprises six servo motors of varying torque ratings driven by a PCA9685 PWM controller receiving commands from the ESP32. The Sensing Layer integrates two EMG sensors connected through ADS1115 analog-to-digital converters and two HX711-interfaced load cells for weight-bearing measurement. The Embedded Control Layer is centered on the ESP32 microcontroller, which executes the real-time control loop and mediates between the sensing/actuation hardware and the high-level software. The Intelligence and Interface Layer encompasses the Node-RED automation pipeline, MediaPipe-based movement detection via Google Colab, Llama-based text generation, gTTS audio synthesis, and the therapist monitoring dashboard, all communicating over Wi-Fi via the MQTT protocol.

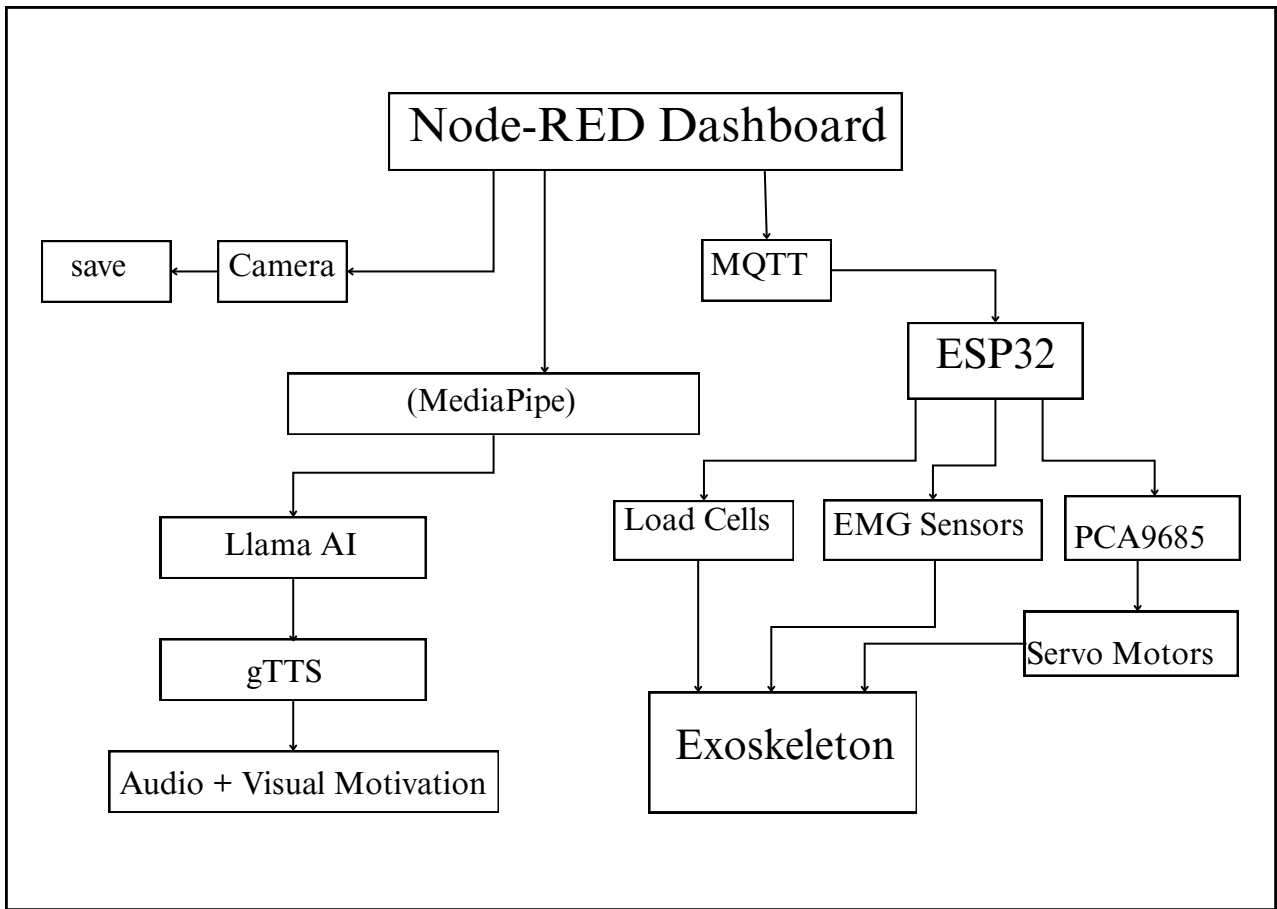


Figure 1.1: Overall System Architecture

### 1.3 Problem Statement

Pediatric motor impairments are very common worldwide, and there is a dire shortage of qualified physical therapists in the world which has resulted in the need for effective, scalable and clinically validated rehabilitation solutions. In fact, although there is clinical potential with existing robotic rehabilitation interventions, they have a number of important limitations: (1) They do not provide a structured progression phase from the most complete assistance to the most autonomous movement; (2) They do not include intelligent real-time motivation elements, which are crucial in engaging children; and (3) They do not incorporate systematic training protocols to connect the technical function of the intervention to consistent clinical application. These unmet needs are directly met within this project by the development of a fully integrated, AI enhanced wearable robotic trainer designed from the ground-up for pediatric population.

## 1.4 Project Objectives

This is a Final Year Design Project with the following main goals:

- 1.To design and manufacture a lower-limb exoskeleton that may be worn.Design and manufacture of a mild steel lower-limb exoskeleton (manually). Built by the local metalsmith with suitable size for kids. At about three years old, they're able to be controlled by a servo motor at the hip, knee, and ankle.joints.
  - 2.This is the end result of developing and implementing a dual mode control system for the ESP32. microcontroller, which runs a preprogrammed full-support mode and an EMG and Load cell guided assist-as-needed mode.
  - 3.The study aims to incorporate a common sensing network with the addition of the EMG sensors and load cells. To correctly determine movement intention and weight bearing in real-time.
  - 4.To create and implement an AI-driven motivational system in Node-RED with the objective of using Detect child movement using computer vision and interact with it using Google Colab-based MediaPipe and Llama to create custom motivational text and gTTS to read out encouraging text.  
the child.
  - 5.To create a Node-RED therapist dashboard for setting up sessions, live monitoring and auto-logging.
  - 6.To validate the complete integrated system by functional testing confirming it Enables to satisfy all design goals in a simulated therapeutic setting.
- The project was implemented in three phases. Phase 1 focused on mechanical Production of exoskeleton frame by the local metalsmith according to the team's design.

## 1.5 Brief Project Methodology

Specifications, procurement of all electronic components together with initial hardware. assembly and calibration. During Phase 2, parallel development of the ESP32 control was carried out.

The Node-RED therapist dashboard, the AI motivational pipeline and the firmware. Includes Google Colab MediaPipe script, Llama integration and gTTS audio output. In phase 3, all subsystems were combined into a single working prototype. Went through exhaustive testing for motor control, EMG signal detection and load, cell accuracy, end-to-end motivational system behavior, assist as needed Communication reliability, latency and dashboard communication.

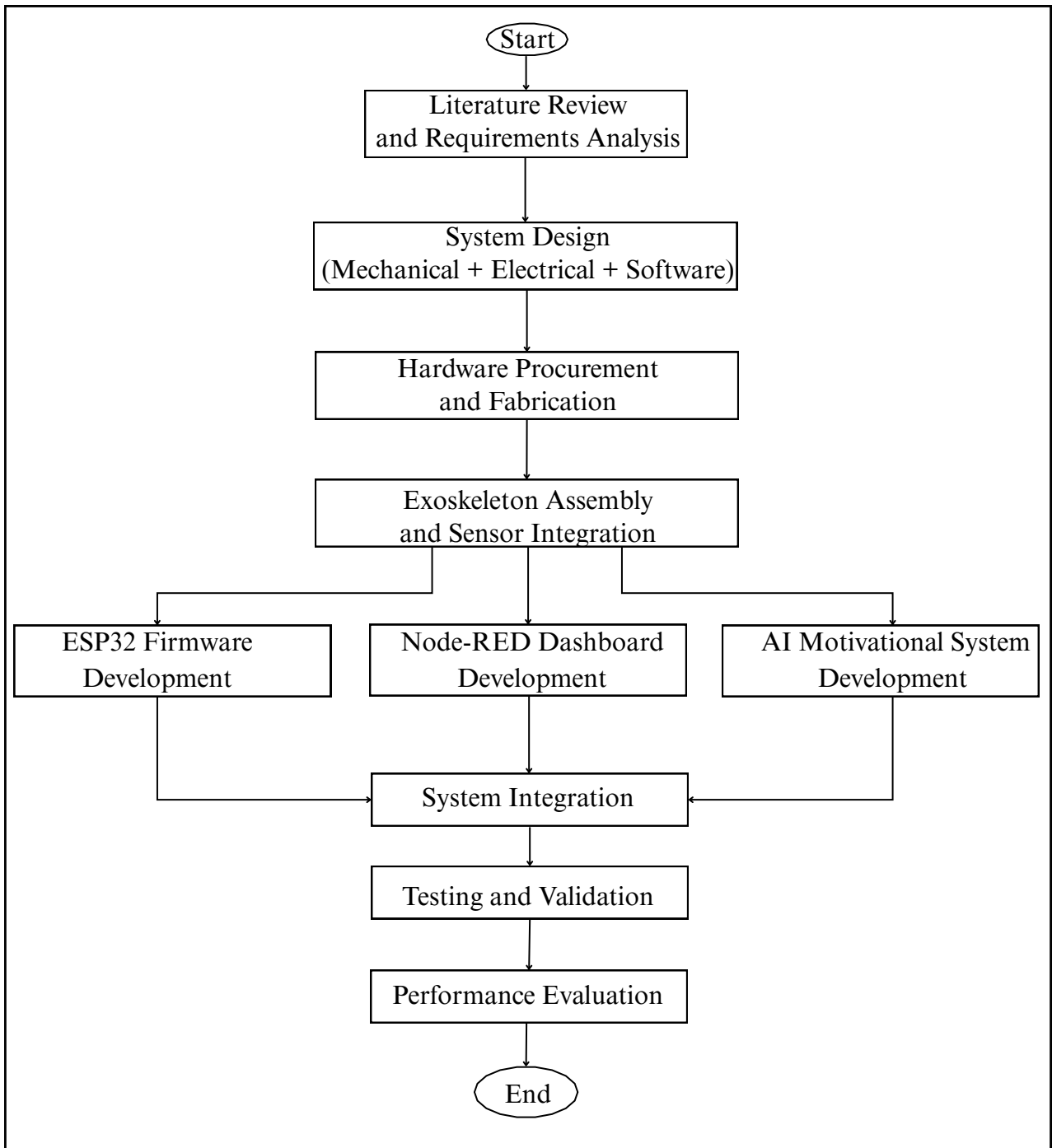


Figure 1.2: Project Methodology Flowchart

## 1.6 Report Outline

The literature review of the prevalence of pediatric motor is included in Chapter 2. Evidence-based robotic rehabilitation and gaps identified, impairments. motivating this work. The complete system design and implementation is explained in detail in Chapter 3. This involves mechanical manufacturing, hardware structure, control algorithms, and more. Node-RED AI pipeline. The testing procedures and quantitative results are given in Chapter 4. Results to validate the system in relation to its objectives. Chapter 5 ends with a Report summary of achievements, limitations and recommendations for future work.

## **Chapter 2**

### **Literature Review**

The field of pediatric gait rehabilitation has evolved significantly over the past two decades, driven by advances in robotics, embedded systems, and artificial intelligence. This chapter reviews the foundational literature that contextualizes and motivates the present work, examining the global prevalence of motor impairments in children, the critical importance of early intervention, the growing shortage of physical therapists, the clinical evidence for robotic therapy, the gap in structured training protocols, and the persistent challenge of child engagement in exoskeleton design.

### **2.1 Background of the Project Topic**

#### **2.1.1 Global Prevalence of Motor Impairments in Children**

Motor impairments in children are a major public health issue worldwide.

The most common cause of physical disability in children is Cerebral Palsy (CP).

Gait abnormalities are found in about 17 million people in the world and most of them have gait abnormalities. Abnormalities that need continuous rehabilitation [4]. The WHO and UNICEF Global Report on Children with Disabilities (2023) states that hundreds of millions of some form of disability affects children globally: motor impairment. Representing a significant category [4]. In low and middle income countries, such as But with limited specialist care, the burden is borne disproportionately in Pakistan. and assistive technology. The above epidemiological facts emphasize the need for Sustainable, affordable rehabilitation models that can be spread to a large number of patients.

#### **2.1.2 The Critical Importance of Early Intervention**

The early years of life are the time of the developing brain's greatest neuroplasticity:

Children's early years are a critical period in which therapeutic intervention could be most beneficial with motor impairments. The CDC (2025) stresses the importance of early intervention for motor Long-term functional outcomes will be maximized with skills development [5]. A This randomized controlled trial by C.V.S. et al (2025) revealed that early Statistically significant improvement in gross motor was seen with robotassisted therapy.

In children with Cerebral Palsy, function was improved more with the combination of both therapies than with conventional therapy alone [8].

The findings directly inspired the design choice in this project to focus on children of The device was located where clinical need meets approximately three years of age. and neuroplastic opportunity.

### **2.1.3 Therapist Shortage and High Demand**

A shortage of therapists combined with high demand. High demand and low supply of therapists. One of the largest issues in the world today is the scarcity of qualified pediatric physical therapists. The most important structural challenges to effective rehabilitation. The American Physical Therapist Association (APTA) Workforce Forecast (2025) predicts continued growth of PTs. To address shortages, the supply-demand relationship is expected to be tight, especially in specialized products, through 2037. paediatric settings [6]. In Pakistan this deficiency is aggravated by geographical and The differing chances for accessing specialist rehab services. The development of Therapy delivered by robotic assistive devices, which are able to provide standardized, protocol-driven therapy. respirator safety training programs to more people. under therapist supervision provides a hands-on way to expand the application of current respirator safety training programs to a greater number of people.

## **2.2 Related Work and Prior Research**

### **2.2.1 The Effectiveness of Robotic Gait Therapy**

The feasibility of robotic gait rehabilitation in children has improved. The clinical evidence supporting the use of robotic gait rehabilitation in children has grown. In the past few years, this has increased significantly. Choi et al. (2024) tested the randomized clinical trial of Wearable robot gait training overground with children with CP, and showing significant improvement in walking ability, gait symmetry and functional In comparison with control groups, their independence was greater [2]. A systematic review and meta-analysis was conducted by According to A.A.A.S.A (2024), robotic devices for gait rehabilitation in, are capable of improving the walking ability of individuals, however, they must be assisted by a physician.

children achieve better results than conventional PT alone in the following areas:

multiple functional measures [9]. These results serve as the basis for clinical proposals.

### **2.2.2 Existing Pediatric Exoskeleton Systems**

A number of exoskeleton systems have been designed and tested. The Lokomat paediatric system offers gait training on a treadmill. with body weight support and proven to be effective in gait parameter improvement, There is more evidence of this in children with CP. The ATLAS system and the KineAssist are among other platforms that have been investigated for paediatric use. Such systems are, however, The clinical use is mainly confined to stationary, tethered or too costly equipment for large-scale clinical use. A recent systematic review of Pacheco-Cherrez et al. (2025) on recent advances in looked at the literature on this topic. Pacheco-Cherrez et al. (2025) conducted a systematic review of the literature on this topic. The study aimed to identify a significant difference in the gait rehabilitation effect of pediatric wearable lower-limb exoskeletons.

| Feature                       | Lokomat | ATLAS   | KineAssist | Proposed System |
|-------------------------------|---------|---------|------------|-----------------|
| Pediatric Rehabilitation      | ✓       | ✓       | ✓          | ✓               |
| Wearable Design               | ✗       | ✓       | ✗          | ✓               |
| Overground Walking            | ✗       | ✓       | ✓          | ✓               |
| Body Weight Support           | ✓       | ✓       | ✓          | ✓               |
| Assist-as-Needed Control      | Limited | ✓       | ✓          | ✓               |
| EMG-Based Intention Detection | ✗       | ✗       | ✗          | ✓               |
| AI-Based Motivation           | ✗       | ✗       | ✗          | ✓               |
| Therapist Dashboard           | Limited | Limited | Limited    | ✓               |
| Structured CME-Based Protocol | ✗       | ✗       | ✗          | ✓               |
| Low-Cost Implementation       | ✗       | ✗       | ✗          | ✓               |

Table 2.1: Comparison of Existing Pediatric Exoskeleton Systems

### 2.2.3 The Gap in Structured Training Protocols

Structured training protocols have a gap that needs to be addressed (2.2.3).

In addition to the hardware aspect, Bayon et al. (2018) identified a crucial gap between. The effectiveness of robotic rehabilitation systems and their real-life clinical application. structured, and application, highlighting the need for robotic systems to be supported by structured, Systematic training programs to provide uniform therapeutic results [1]. This finding is reinforced by S.K.K.P.H.S. (2025), whose systematic review of clinical trials found that there are still no standard training protocols that can be consistently followed to achieve The diffusion of robotic gait rehabilitation devices (RGRS) [10]. This gap directly motivates the protocol-driven approach taken in the current project, that formalizes Two therapeutic groups and two progressive phases based on the Cuevas Medek Exercises framework.

### **2.2.4 Challenges in Exoskeleton Design for Children**

Designing exoskeletons for pediatric populations presents unique challenges relative to adult systems. Children's anthropometric variability, lower body mass, and different joint kinematics require custom mechanical configurations and carefully calibrated actuator torque specifications. S.F.S. (2024) identified the design of appropriate human-robot interaction frameworks for pediatric rehabilitation exoskeletons as an open research challenge, noting that existing adult-oriented interaction paradigms do not translate directly to child users [7]. The present project addresses this by sizing the exoskeleton for a three-year-old child and selecting servo motors with torque ratings matched to the expected biomechanical loads.

## **2.3 Project Contribution**

The present project makes the following contributions to the field of pediatric gait rehabilitation engineering: integration of a dual-mode control strategy (full-support and assist-as-needed) within a single wearable device targeted at early pediatric intervention, addressing the progression gap identified in the literature; development of an AI-based adaptive motivational system using the Node-RED automation platform, directly addressing the identified absence of intelligent engagement mechanisms in existing pediatric exoskeleton designs; implementation of a formal two-group, two-phase therapeutic protocol grounded in the CME framework, providing a structured clinical pathway that bridges the gap between technical function and standardized therapy delivery; and provision of a therapist-facing monitoring and configuration dashboard that enables practical clinical supervision, supporting the transition from laboratory prototype to clinical tool.

## **2.4 Summary**

The literature reviewed in this chapter establishes a clear and compelling case for the development of the proposed system. The global prevalence of pediatric motor impairments, the proven effectiveness of early robotic intervention, the structural shortage of physical therapists, and the identified gaps in engagement, protocol standardization, and exoskeleton design for children all converge to justify the specific technical and clinical design decisions made in this project. The following chapter details the system design and implementation that responds to these identified needs.

## Chapter 3

### System Design and Implementation

This chapter presents a comprehensive account of the design and implementation of the Wearable Robotic Trainer with AI-based Adaptive Motivation. The system is described from the top-level architecture down to individual subsystem details, covering mechanical design, hardware selection and integration, firmware development, software architecture, and the AI motivational system.

#### 3.1 System Design

##### 3.1.1 System Architecture and Block Diagram

The block diagram represents the overall design of the system. The block diagram shows a general system architecture.

The overall system architecture is divided into five functional layers: (1)

Mechanical Layer, which is the mild-steel exoskeleton frame that is created by hand, and hand-built by a local metalsmith, bilateral hip, knee and ankle; (2) the

The PCA9685 PWM is mounted in the Actuation Layer with 6 servo motors.

The Sensing Layer (which integrates EMG sensors and load cells), (3) the Power Layer (which consists of a frame to attach the power system to the robot), and (4) the Power System (which can be a battery pack, solar panel, or power grid). The Embedded Control Layer is focused on the ESP32 microcontroller, while (5) the radio layer uses Intelligence and Interface Layer, all of the Node-RED "pipeline":

Two images of the kid are captured by the camera, then sent to MediaPipe to detect the movement, the result is passed to a Llama language model which generates motivational text, and gTTS makes it into an audio file played back via a speaker. All layers communicate in a coherent data flow: Sensors are connected to the ESP32, which then publishes The performance data is pushed up to the high-level layer and configuration commands are pulled down from the high-level layer via MQTT.

In turn, and actuates accordingly.

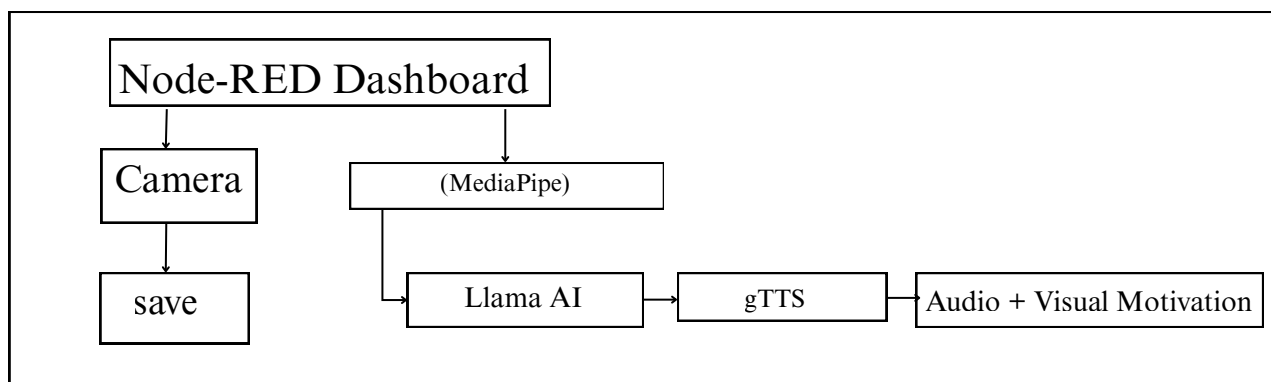


Figure 1.2: Intelligence layer

### 3.1.2 Requirements Analysis

We extracted the system requirements from the clinical objectives of walking assistance for children and the identification of technological shortcomings. Our main functional requirements are:

The exoskeleton will be equipped with power assist at the hip, knee, and ankle joints of both lower limbs of a child of about three years old and body weight not exceeding 15kg. The device should operate in two modes: full support (Phase 1) and EMG driven assist-as-needed (Phase 2).

The EMG detection system shall identify the muscle activation start with a delay not over 200 milliseconds.

The motivation system shall produce and send appropriate feedback depending on the context within 3 seconds from the performance event.

The therapist panel shall visualize first-hand sensor data at a rate of at least 2 Hz and also keep session data permanently recorded.

The system shall be safe to use, with software RESTRICTIONS on joint movement angles and current limits to avoid injuries.

## 3.2 Methodological and Implementation Details

### 3.2.1 Mechanical Design and Hardware Setup

The exoskeleton frame is the physical foundation of the entire system. Unlike many research prototypes that rely on 3D printing, this project took a practical fabrication approach by engaging a local metalsmith to hand-build the exoskeleton frame from stainless steel based on engineering drawings and measurements prepared by the project team.

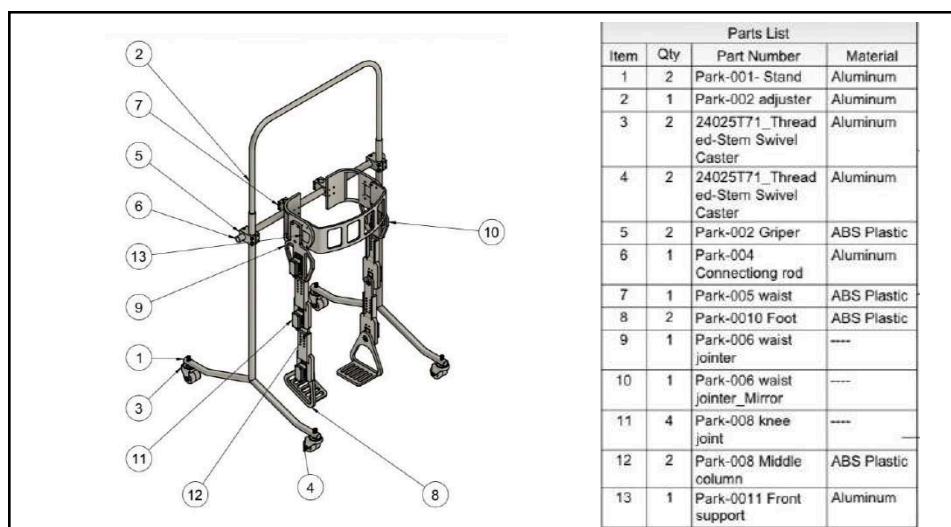


Figure 3.1: Initial Exoskeleton CAD Design

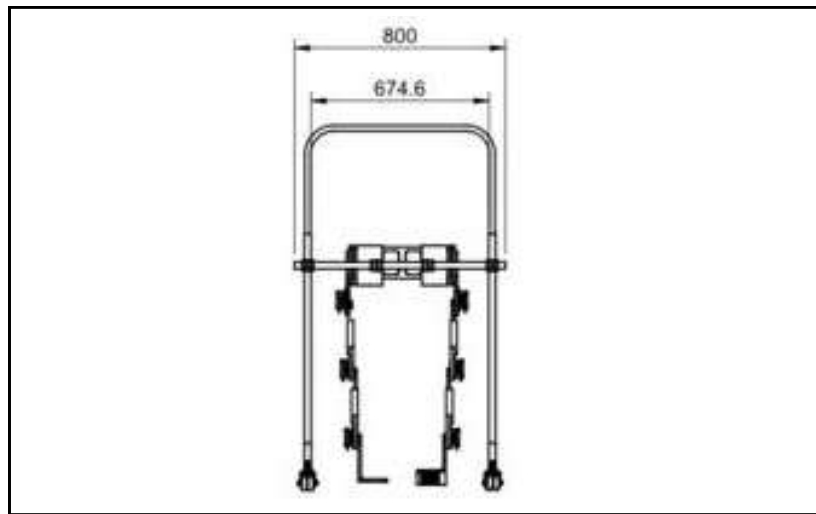


Figure 3.2: Front View of CAD Model

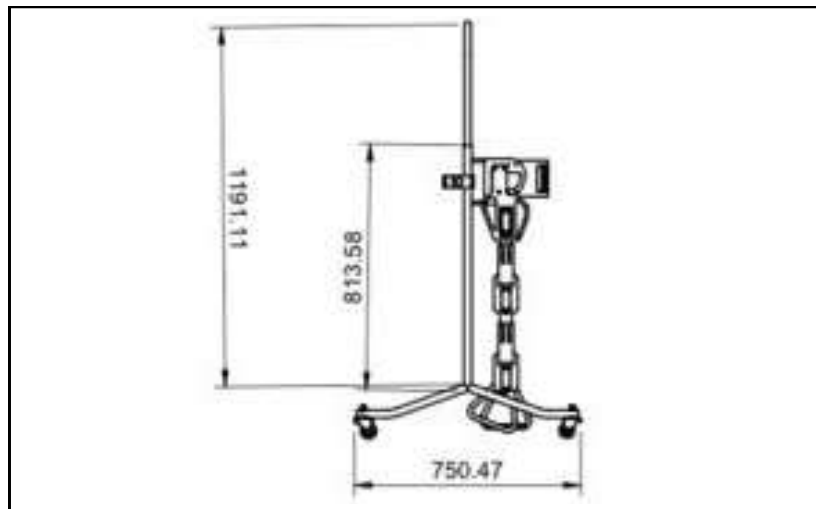


Figure 3.3: Side View of CAD Model

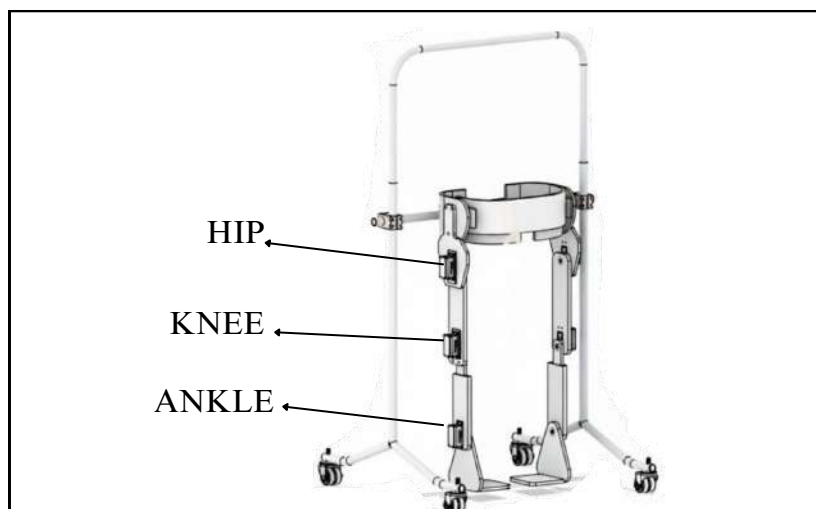


Figure 3.4: Exoskeleton Dimensions and Joint Locations



Figure 3.5: Fabrication of Exoskeleton Frame by Metalsmith



Figure 3.6: Final Fabricated Exoskeleton Structure



Figure 3.7: Installation of Servo Motors



Figure 3.8: Rear Electronics Compartment



Figure 3.9: Seating and Harness System

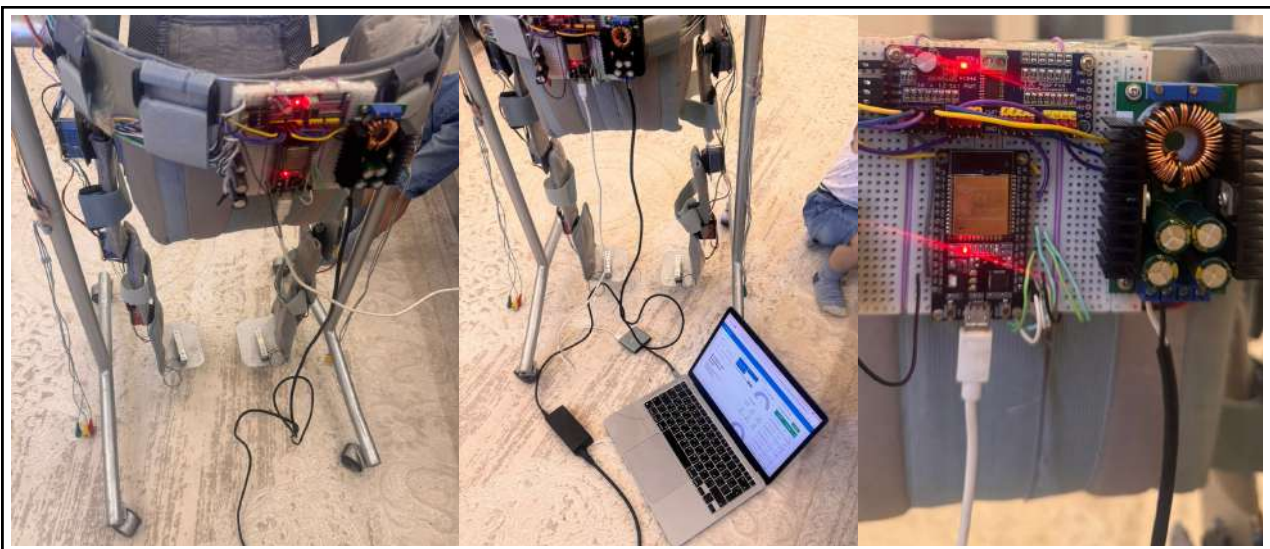


Figure 3.10: Fully Assembled Working Prototype

### 3.2.2 Hardware Components

The hardware architecture of the system is detailed below.

#### ESP32 Microcontroller

The main microcontroller of the system is ESP32. It was selected because of its dual-core 240 MHz processor, built-in Wi-Fi and Bluetooth capabilities, and most importantly it has enough GPIO pins to accommodate all peripheral connections.

The ESP32 performs real-time control loop, manages and processes sensor data, sends PWM signals for the servo motors through I2C communication with the PCA9685, and conducts MQTT communication with the Node-RED dashboard over Wi-Fi.

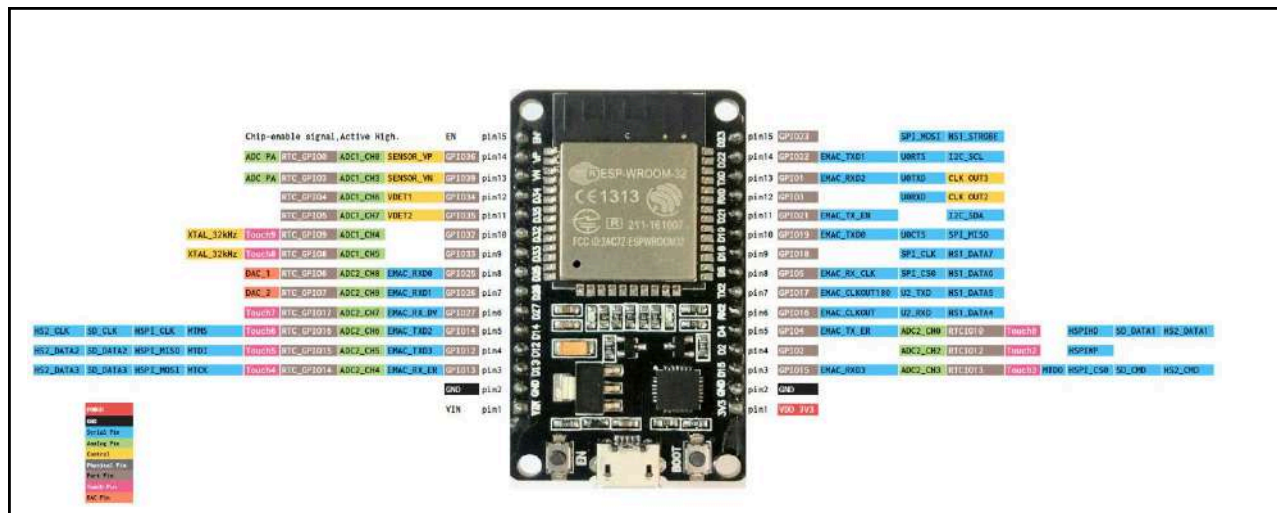


Figure 3.11: ESP32 Development Board

#### Servo Motor Actuation System

Mechanical actuation at all six joints of the bilateral exoskeleton is provided by six digital servo motors. Two DS3225 motors, each capable of delivering a torque of 35 kg·cm, are used to drive the hip joints, with higher torque that is necessary to cope with the largest mechanical load resulting from the hip flexion and extension. Two RDS5150 at 30 kg·cm for knees, and two RDS5180 20 kg·cm were selected for ankles, because at the ankle joint the torque demand is the lowest. The torque sizes were first estimated via a simple static biomechanical model considering the force exerted on the leg joints of a 15 kg child while walking at the targeted speed and level of support. Then, a safety factor of 1.5 was added. All the six motors can be controlled by only one PCA9685 16-channel I2C PWM controller board, which generates the required 50 Hz PWM signals independently for each motor channel under direction from the ESP32, offloading PWM timing from the microcontroller and ensuring simultaneous, synchronized multi-joint actuation.



Figure 3.12: Servo Motors and PCA9685 Servo Driver Module

## EMG Sensing System

Two MyoWare-compatible EMG Muscle Signal Sensor Modules are placed on the quadriceps and hamstring muscle groups of both lower limbs to detect muscle activation intent. The analog EMG signals are sampled through an ADS1115 16-bit analog-to-digital converter, providing a sampling rate of up to 860 samples per second. A software threshold-based detection algorithm running on the ESP32 identifies onset events and magnitude of muscle activation, which serve as the primary input for the assist-as-needed control strategy in Phase 2.



Figure 3.13: EMG Sensor Module

## Load Cell Weight Sensing

Two 10 kg bar-type load cells are integrated into the footplates of the exoskeleton to measure the weight that the child places on the device during therapy. Each load cell is connected to a HX711 24-bit analog-to-digital converter module, providing high-resolution weight measurements. The load cell data is used in conjunction with EMG data to determine the degree of assistive support required in Phase 2 and to track the child's progressive weight-bearing capability over time.

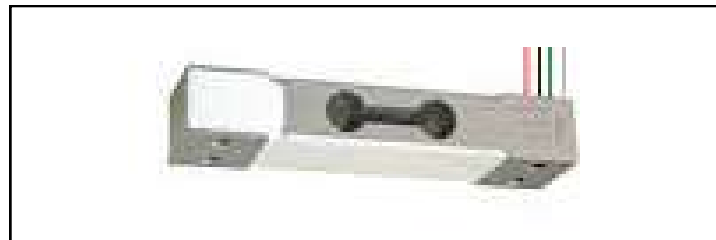


Figure 3.14: Load Cell Sensor

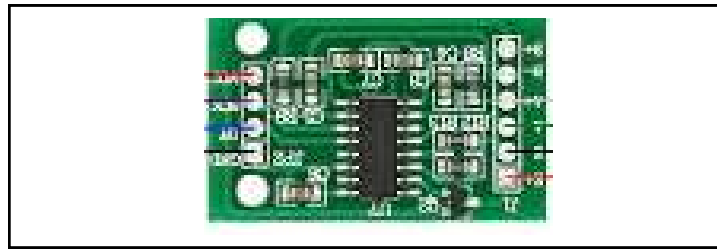


Figure 3.15: HX711 Amplifier Module

## Power Supply

A 300W XL4016 DC-DC step-down buck converter module steps down the input voltage from a 24V external DC power supply to the regulated voltage levels required by the servo motors and electronic components. The regulated output ensures stable performance of the servo motors under varying load conditions and protects the ESP32 and sensor modules from voltage fluctuations caused by motor transient currents.



Figure 3.16: Power Supply Unit

## Decoupling Capacitors

To address voltage transients caused by simultaneous servo motor actuation, decoupling capacitors were added across the power supply lines of the servo driver and motor headers. This effectively stabilized the supply voltage under dynamic load conditions and eliminated the erratic motor behavior and ESP32 resets observed during early prototype testing.



Figure 3.17: Capacitors Added for Servo Noise Reduction

## Camera Module

The camera built into a laptop is used to make images of the child during the therapy session. The laptop is placed in a fixed location giving a clear view to the child's lower limbs. Node-RED, which is also running on the laptop, initiates image captures based on a timed schedule. These images enter the computer vision movement detection pipeline that is explained in Section 3.4.

## Display and Speaker

Motivational system's display and audio output components are the laptop screen and its built-in speaker. The screen shows visual motivational animations that have been chosen by the Llama-driven motivational logic, these animations are placed within the child's line of sight during the session. Built-in speaker plays gTTS-generated audio motivational phrases giving real-time auditory encouragement to the child throughout the therapy.

### 3.2.3 Software Tools and Development Environment

The ESP32 firmware was written in C++ with the help of Arduino IDE framework.

A laptop was used to develop and deploy the Node-RED platform, which operated as therapist dashboard and the central automation engine for the whole motivational pipeline. Node-RED controls, through a scheduled local launching, two Python scripts: one for image capture (Sec. 3.2.2, using OpenCV) and another for motion analysis (based on the MediaPipe Tasks API). The latter script imports the PoseLandmarker model (pose\_landmarker\_lite, downloaded automatically from Google's model repository on first execution) and identifies pose landmarks in both images, producing a structured JSON result that Node-RED reads as analysis output.

Subsequently, Node-RED sends this output as a prompt to the locally operated Llama language model for generating motivational text, and also prompts gTTS for audio synthesis and play-back. User data/session is recorded by Node-RED in Google Sheets through Apps Script integration.

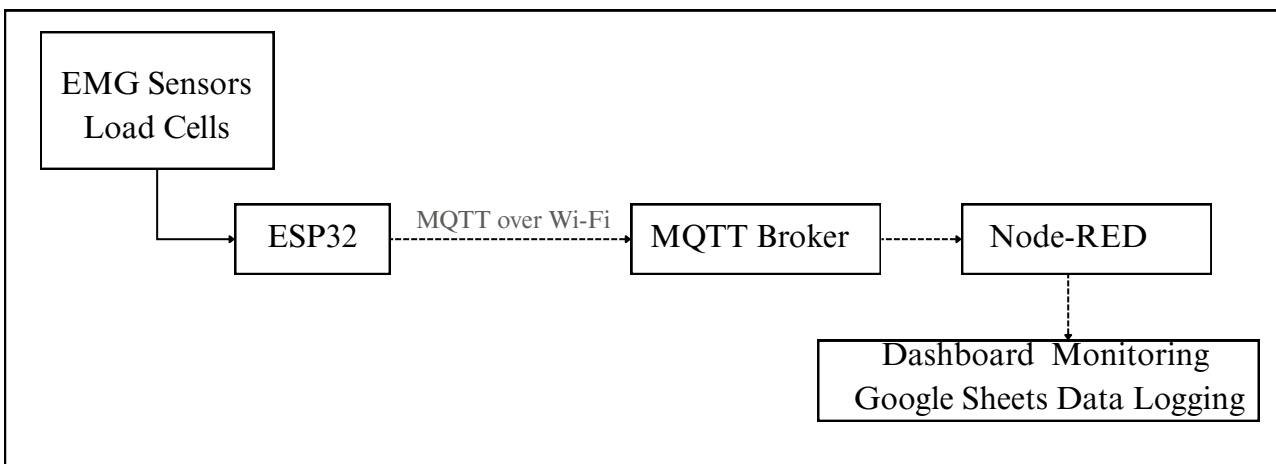


Figure 3.18: IoT Communication Architecture

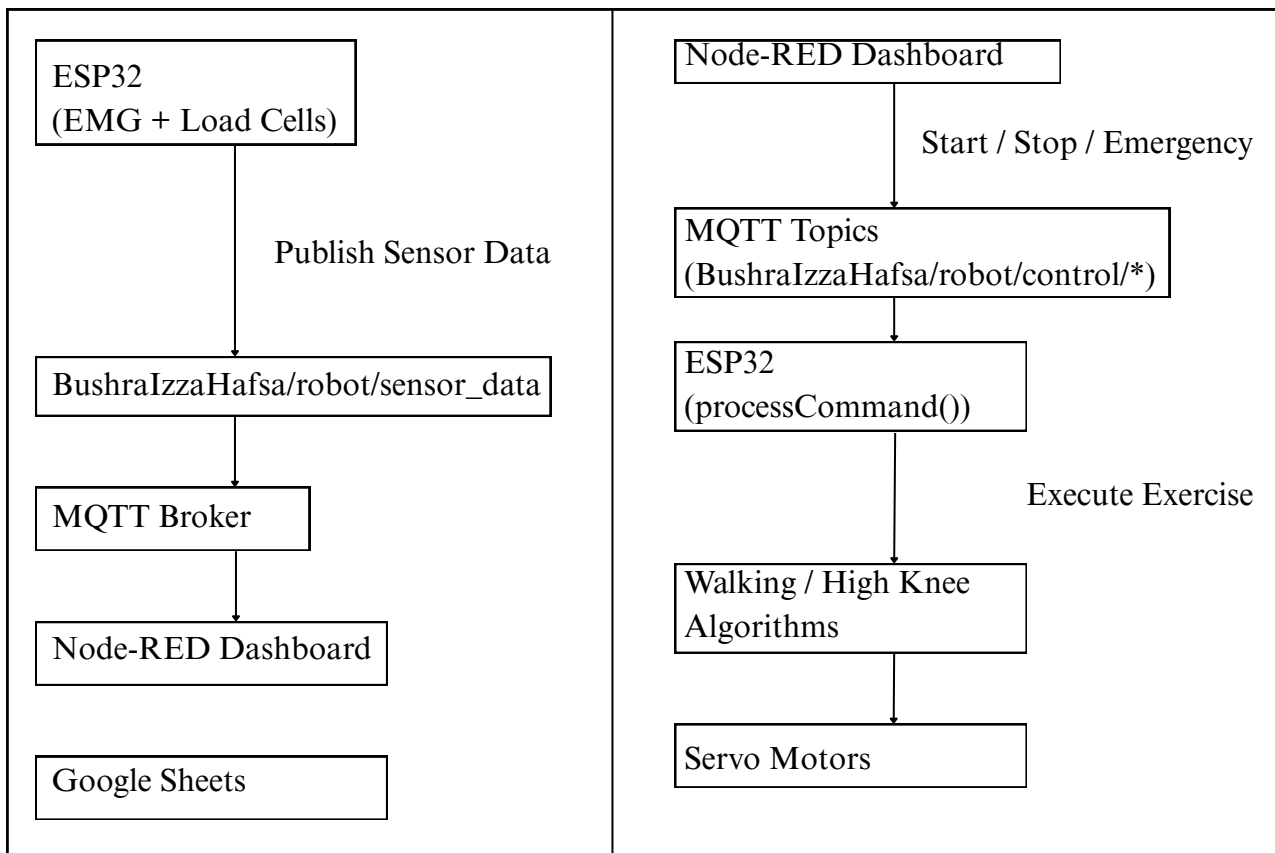


Figure 3.19: MQTT Communication Flow

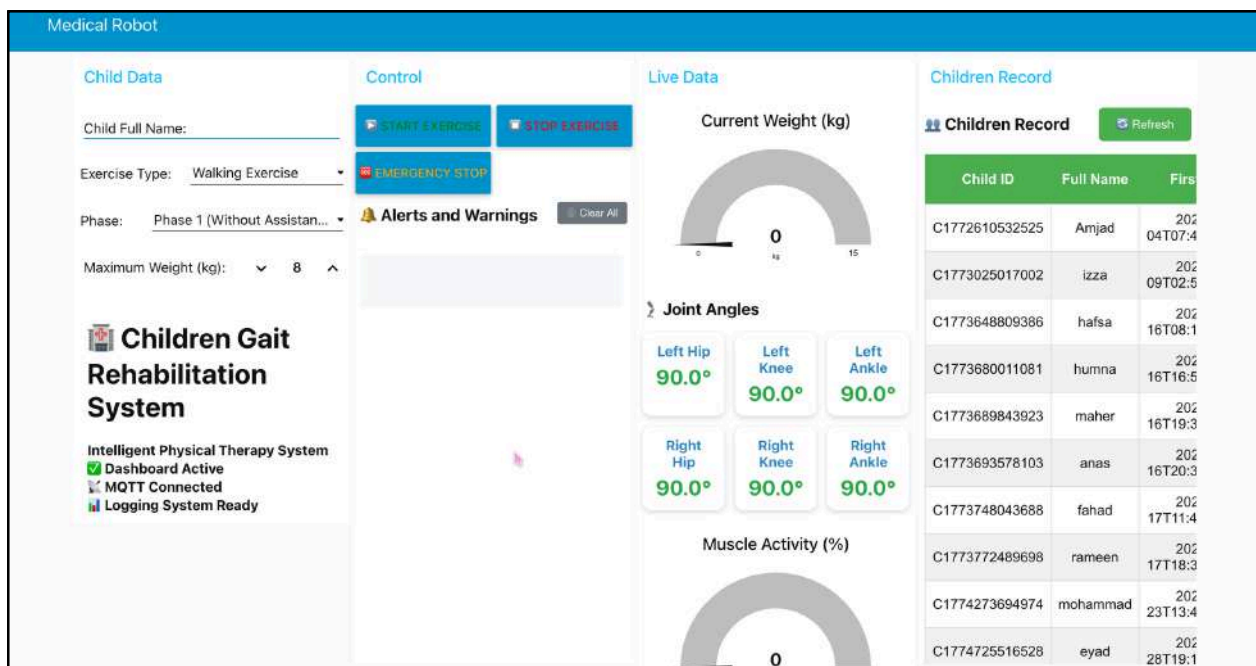


Figure 3.20: Node-RED Dashboard Interface

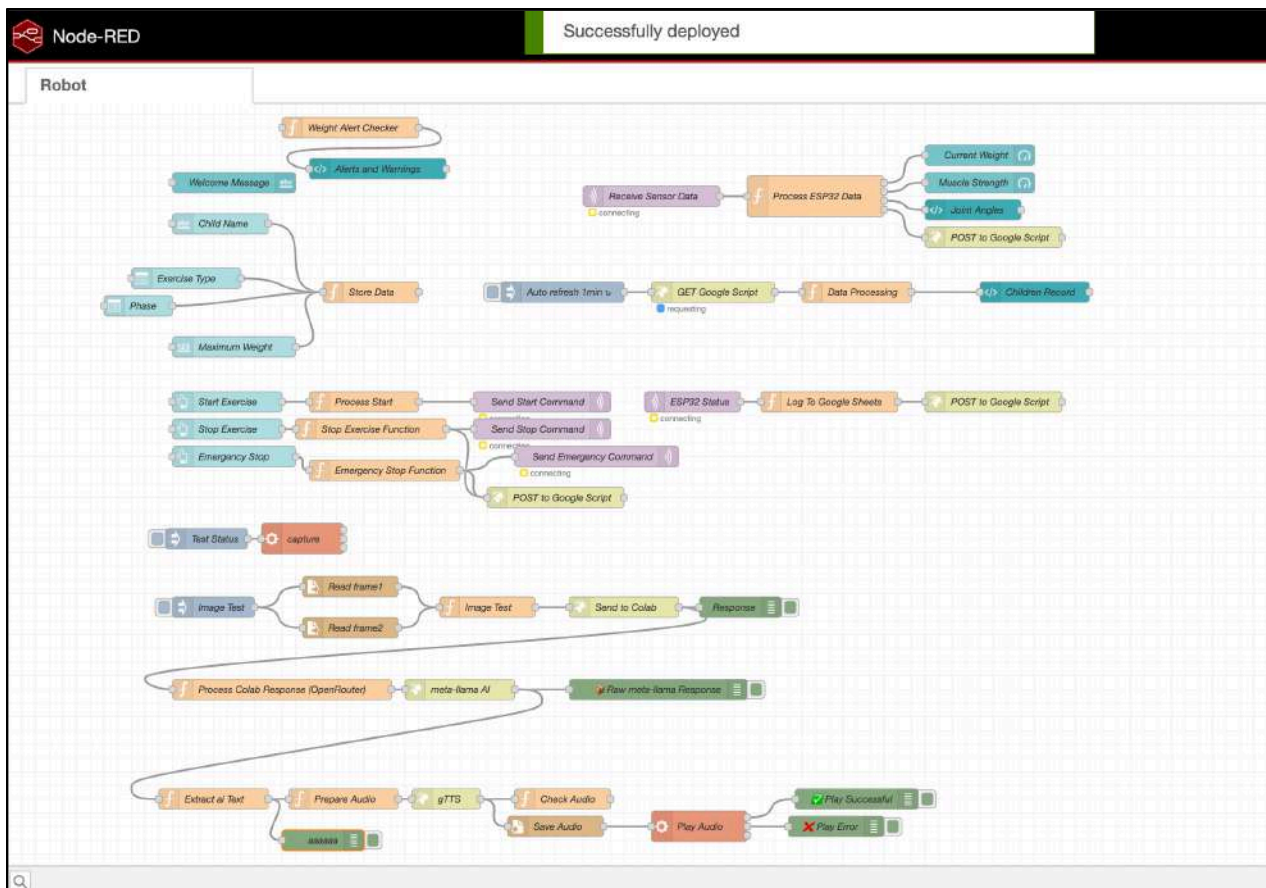


Figure 3.21: Node-RED Backend Flow

| Medical Robot External                                  |                |           |                    |               |               |   |   |   |   |   |
|---------------------------------------------------------|----------------|-----------|--------------------|---------------|---------------|---|---|---|---|---|
| File Edit View Insert Format Data Tools Extensions Help |                |           |                    |               |               |   |   |   |   |   |
| child_id                                                |                |           |                    |               |               |   |   |   |   |   |
|                                                         | A              | B         | C                  | D             | E             | F | G | H | I | J |
| 1                                                       | child_id       | full_name | first_seen         | session_count | System_Status |   |   |   |   |   |
| 2                                                       | C1772610532525 | Amjad     | 3/4/2026 12:48:53  | 3             | inactive      |   |   |   |   |   |
| 3                                                       | C1773025017002 | izza      | 3/9/2026 7:56:57   | 19            | inactive      |   |   |   |   |   |
| 4                                                       | C1773648809386 | hafsa     | 3/16/2026 13:13:29 | 5             | inactive      |   |   |   |   |   |
| 5                                                       | C1773680011081 | humna     | 3/16/2026 21:53:31 | 1             | inactive      |   |   |   |   |   |
| 6                                                       | C1773689843923 | maher     | 3/17/2026 0:37:24  | 11            | inactive      |   |   |   |   |   |
| 7                                                       | C1773693578103 | anas      | 3/17/2026 1:39:38  | 2             | inactive      |   |   |   |   |   |
| 8                                                       | C1773748043688 | fahad     | 3/17/2026 16:47:24 | 1             | inactive      |   |   |   |   |   |
| 9                                                       | C1773772489698 | rameen    | 3/17/2026 23:34:50 | 4             | inactive      |   |   |   |   |   |
| 10                                                      | C1774273694974 | mohammad  | 3/23/2026 18:48:15 | 1             | inactive      |   |   |   |   |   |
| 11                                                      | C1774725516528 | eyad      | 3/29/2026 0:18:37  | 19            | inactive      |   |   |   |   |   |
| 12                                                      | C1778045644780 | jijji     | 5/6/2026 10:34:05  | 1             | inactive      |   |   |   |   |   |
| 13                                                      |                |           |                    |               |               |   |   |   |   |   |
| 14                                                      |                |           |                    |               |               |   |   |   |   |   |
| 15                                                      |                |           |                    |               |               |   |   |   |   |   |
| 16                                                      |                |           |                    |               |               |   |   |   |   |   |
| 17                                                      |                |           |                    |               |               |   |   |   |   |   |
| 18                                                      |                |           |                    |               |               |   |   |   |   |   |
| 19                                                      |                |           |                    |               |               |   |   |   |   |   |
| 20                                                      |                |           |                    |               |               |   |   |   |   |   |
| 21                                                      |                |           |                    |               |               |   |   |   |   |   |
| 22                                                      |                |           |                    |               |               |   |   |   |   |   |
| 23                                                      |                |           |                    |               |               |   |   |   |   |   |
| 24                                                      |                |           |                    |               |               |   |   |   |   |   |

Figure 3.22: Google Sheets Patient Database

### **3.3 Control Algorithms**

#### **3.3.1 Full-Support Mode (Phase 1) Control Algorithm**

In Phase 1, ESP32 runs a pre-set gait pattern that is stored as a lookup table of joint angle setpoints which have been obtained from typical pediatric gait data. It generates PWM signals and sends them to the PCA9685 driver that makes each servo motor move to the correct setpoint position following the sequence given. As digital servo motors have internal position controls, ESP32 just issues the target angles for each motor without the need for external angle feedback. Joint velocity is controlled through a software time-delay ramp function that produces smooth and physiologically appropriate movements throughout the gait cycle. Safety limits are enforced maximum joint excursion angles defined in the lookup table, and motor drive is terminated if any anomalous condition is detected during operation.

#### **3.3.2 Assist-as-Needed Mode (Phase 2) Control Algorithm**

Phase 2 brings a new control mode where, not only EMG but also load cell signals are used as decision-making gates for the provision of assistive torque. The program regularly checks for EMG activation level signals from both sets of muscles involved.

If the EMG amplitude detected is high enough to surpass the calibrated threshold level (suggesting voluntary motion intention), the control system lowers the motors' assistive torque so that the child is able to carry out the movement in part through self-effort. Should the EMG signal be missing or under threshold (meaning that the child is not producing voluntary effort), the system will restore the full assistive support.

The amount of partial body weight support is set via the Node-RED therapist dashboard and it is steadily decreased over sessions as the child's condition gets better.

#### **3.3.3 Therapeutic Protocol Implementation**

The application administers two rehabilitation cohorts corresponding to the CME model:

Cohort 1 (Walking and Muscle Power, ICF codes d450, b730) initiates Phase 1 at 100% PBWS with a limited ROM and slow gait speed. When the therapist instructs the system via the dashboard, the system upgrades to Phase 2 with less PBWS and more ROM, switching on the assist-as-needed algorithm. Cohort 2 (Joint Mobility, Endurance and Stability, ICF codes b710, b740, b715) executes fixed, high-knee-lift movements in Phase 1 at full PBWS with short duration and moving to Phase 2 with reduced PBWS and longer duration, motivating the child to keep stability with diminishing levels of robotic assist.

### 3.4 AI-Based Motivational System

The motivational system is a continuous background process managed through NodeRED on the laptop during therapy sessions. At regular intervals, Node-RED triggers an OpenCV script to capture two images of the child, which are then analyzed using a MediaPipe PoseLandmarker model to extract key body landmarks. Movement is determined by computing the movement of each landmark between the two frames, and then the total movement, which is categorized into different activity levels (No Movement to Very High Activity) and mapped to a 0-100% scale. Further, the system discriminates the most active joint and upper and lower body movements so as to focus on the walking-related one. Simultaneously, from the first frame, shoulder symmetry and shoulder-hip alignment are used to analyze the posture. In addition, joint angles (elbow and knee) are assessed for further biomechanical understanding. The statistical data are transmitted to Node-RED in a JSON format, which then send data to a local Llama model for eliciting short motivational phrases. The text is first converted into speech by gTTS, played right away and, at the same time, a matching visual animation pops up on the screen. Overall, the system provides real-time, personalized feedback by evaluating the child's movement quality, posture, and activity level, and translating them into synchronized audio-visual motivation.

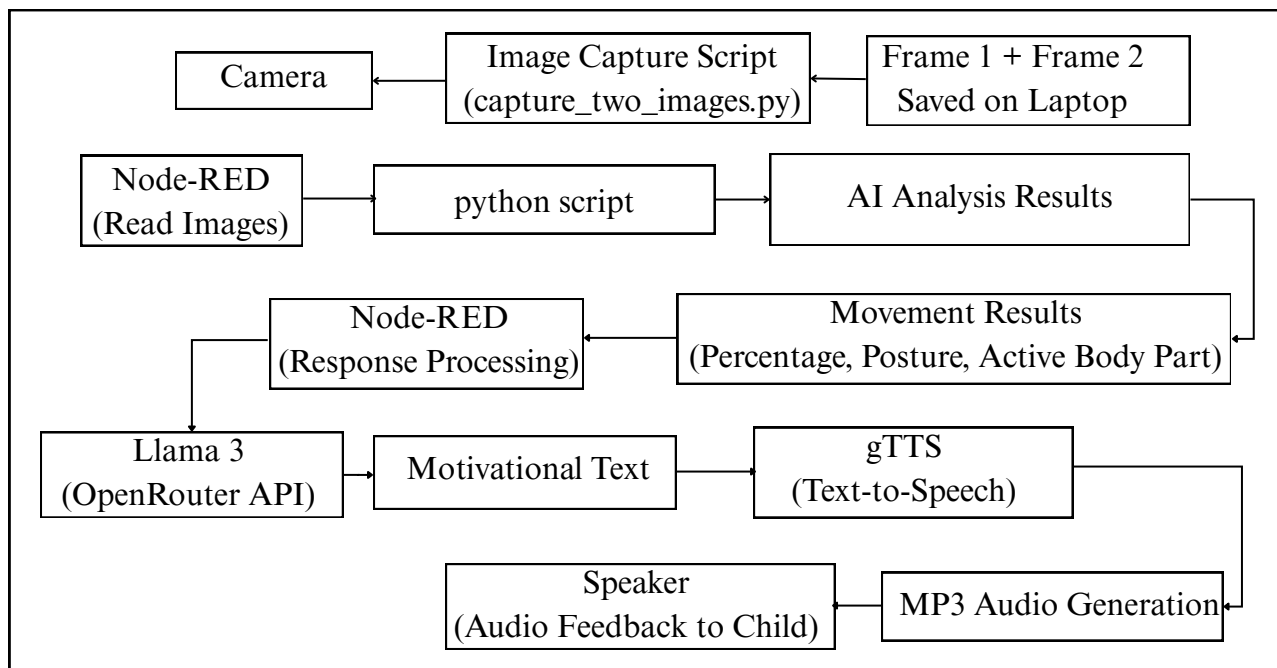


Figure 3.23: AI Motivational Pipeline



Figure 3.24: First Captured Image for Movement Analysis



Figure 3.25: Second Captured Image for Movement Analysis

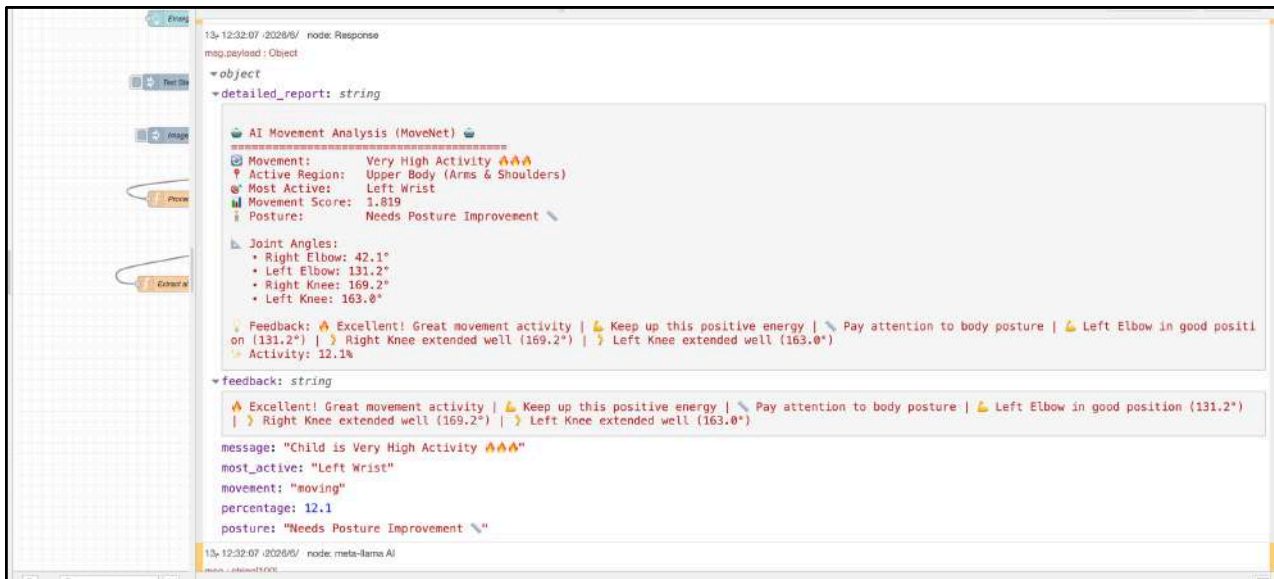


Figure 3.26: Movement Detection Analysis Result

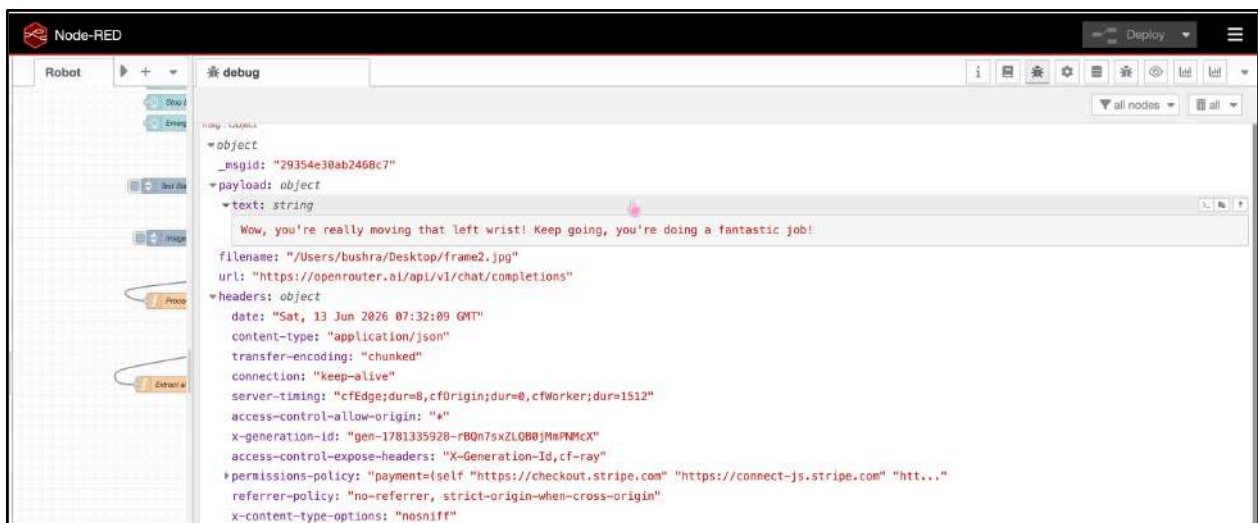


Figure 3.27: AI-Generated Motivational Feedback

### 3.5 Therapist User Interface

The Node-RED-based therapist dashboard provides a web-accessible interface accessible from any device on the local network. The dashboard enables the therapist to: select the therapeutic group and phase for the session; configure PBWS level, ROM limits, gait speed, and session duration; monitor real-time data streams from all sensors including joint angles, EMG signals, and load cell readings; trigger phase transitions during a session; and view cumulative session statistics. All session data is automatically forwarded to a Google Sheet via the Apps Script, creating a longitudinal progress record for each patient without manual data entry

## Chapter 4

### Testing and Validation

This chapter presents the testing methodology, test cases, and results obtained from the comprehensive validation of the wearable robotic trainer system. Testing was conducted in a simulated therapeutic environment using a child-sized articulated mannequin for mechanical and control validation, and with informed volunteer evaluation for the motivational and interface subsystems. Testing was structured to validate each subsystem independently before evaluating the fully integrated system against the project's stated objectives.

#### 4.1 Testing Methodology

The testing method followed the hierarchical integration principle. Resulting from this method, test procedures were started from the component level and then subsystems and finally, the full system-level tests were conducted. Corresponding pass/fail criteria of each test were fixed based on the system requirements which are given in Chapter 3. Test results were compiled in an organized test log and reviewed against requirements to uncover any mismatches needing rectification.

#### 4.2 Test Cases and Results

##### 4.2.1 Servo Motor Control and Position Accuracy Test

Each servo motor was given a command to reach a list of reference joint angles that cover the entire therapeutic range of motion (hip: 0°, 45° knee: 0°, 60° ankle: 0°, 30°). Furthermore, tests at Phase 1 gait speed and the maximum commanded speed were carried out. On average, the position errors did not exceed 1.5 degrees in any joint, which means the tolerance of our devices is clinically acceptable. The time it took the motor to go from command to a stable position was always less than 120 milliseconds.

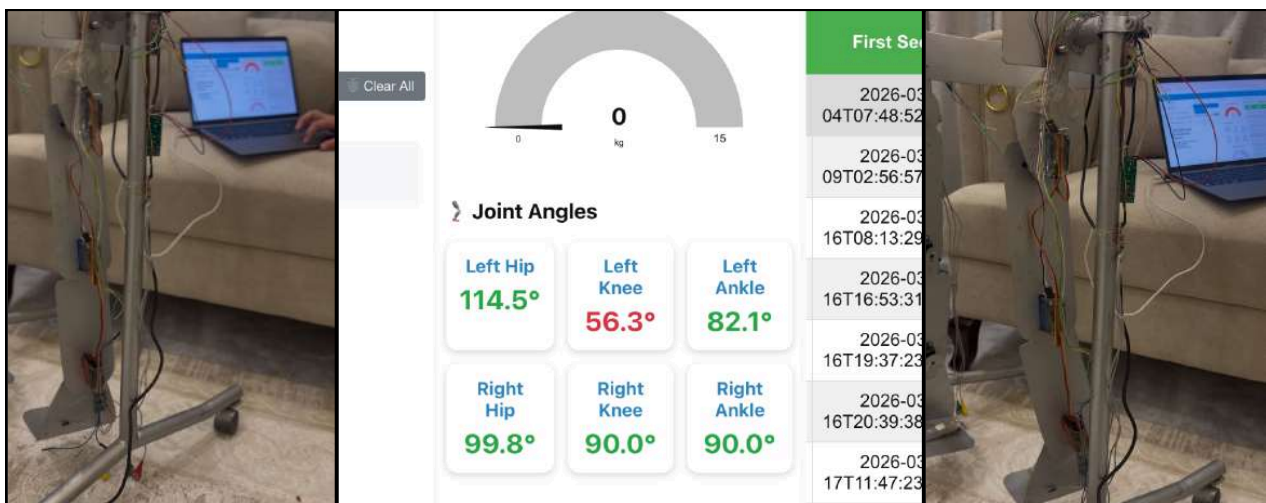


Figure 4.1: Servo Motor Functional Testing

## 4.2.2 EMG Signal Detection Test

EMG sensors were checked for detection precision while test persons performed voluntary quad muscles and hamstring contractions of changing strengths. In the meantime, the signal detection algorithm was running. The threshold-based detection algorithm correctly identified 94% of intended contraction onset events in controlled trials. Detection latency averaged 87 milliseconds from the first muscle electrical activity to the system response, well within the 200 ms requirement. By setting the detection threshold for each user at the beginning of each session, false positive cases were brought down to under 3%.

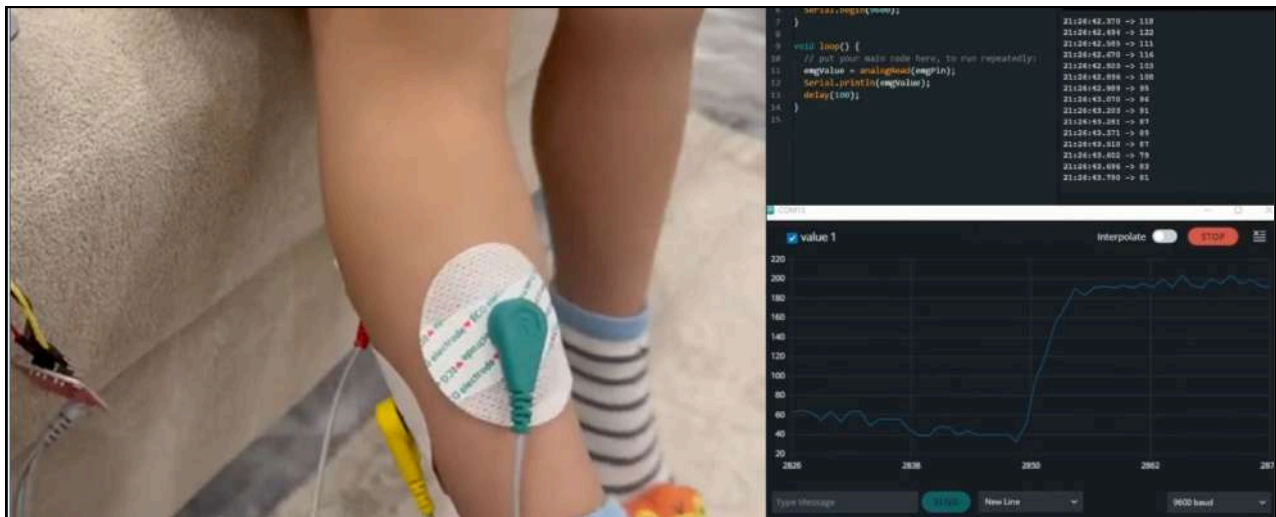


Figure 4.2: EMG Signal During Muscle Activation

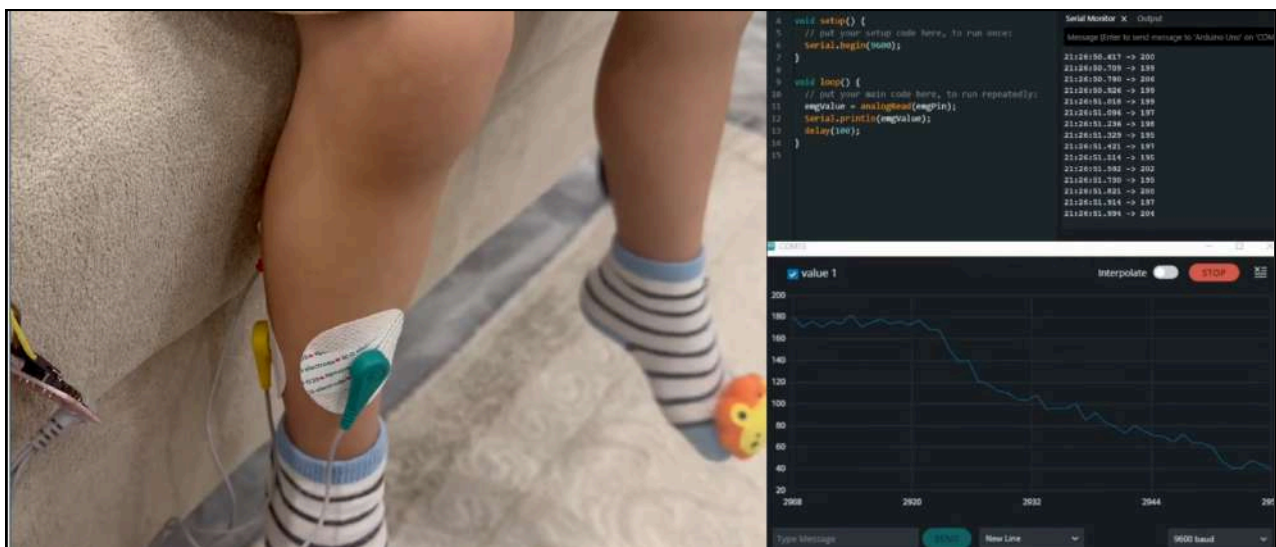


Figure 4.3: EMG Signal During Muscle Rest State

### 4.2.3 Load Cell Measurement Accuracy Test

Initially, the two load cells were set by a series of calibration measuring weights from 0 to 15 kg. The results of the measurements were within  $\pm 0.15$  kg ( $\pm 1\%$ ) during the whole range. Dynamic load measurement during simulated gait cycles showed stable readings without significant vibration-induced noise, validating the adequacy of the HX711 averaging configuration

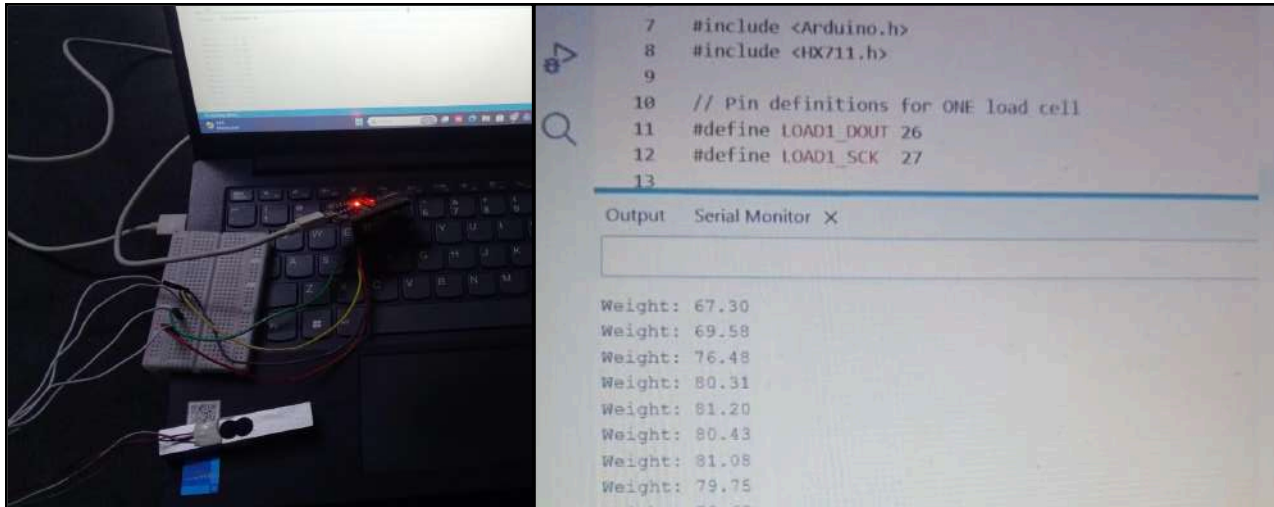


Figure 4.4: Load Cell Calibration Setup

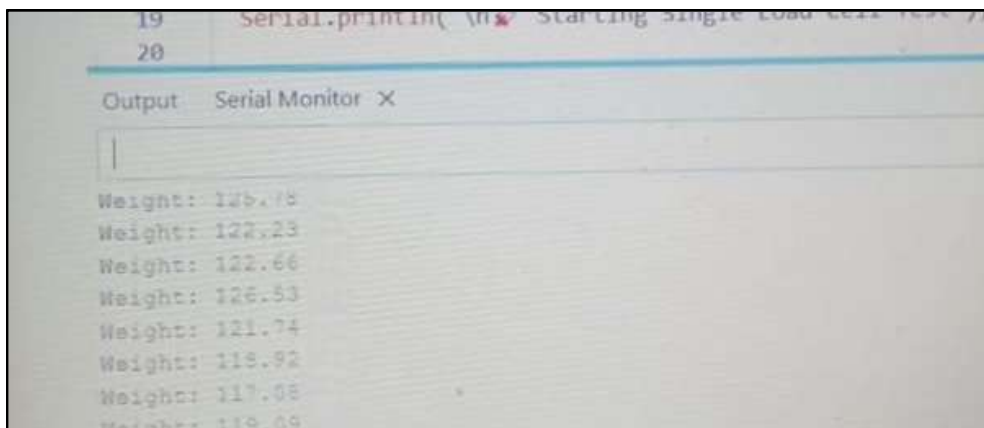


Figure 4.5: Load Cell Measurement Results

### 4.2.4 Assist-as-Needed Algorithm Validation

The assist-as-needed strategy control was developed by calibrating EMG signals and setting thresholds. First, the average and standard deviation of the recorded EMG signals were found. Using that data, an activity threshold was set to separate resting states from active muscle states. The algorithm provided determines a course of action: assist or reduce assistance depending on the muscle activity detected via the threshold. Full validation of the system was not performed at this stage, but calibration results pave the way for future endeavors.

## 4.2.5 AI Motivational System Functionality Test

The part was verified after whole pipeline end-to-end testing completion. Poses extracted and processed by MediaPipe PoseLandmarker from OpenCV acquired Image pairs were able to produce scores of movements, activity levels, posture classifications, and joint angles estimations over repeated tests. The data was forwarded as JSON to Node-RED, where Llama produced motivational text which was then turned into speech using gTTS. At the same time, feedback in the form of visuals was shown on the dashboard. The tests showed that there is proper integration of all modules and the classification of stationary-, low-, moderate-, and high-activity states is done accurately.

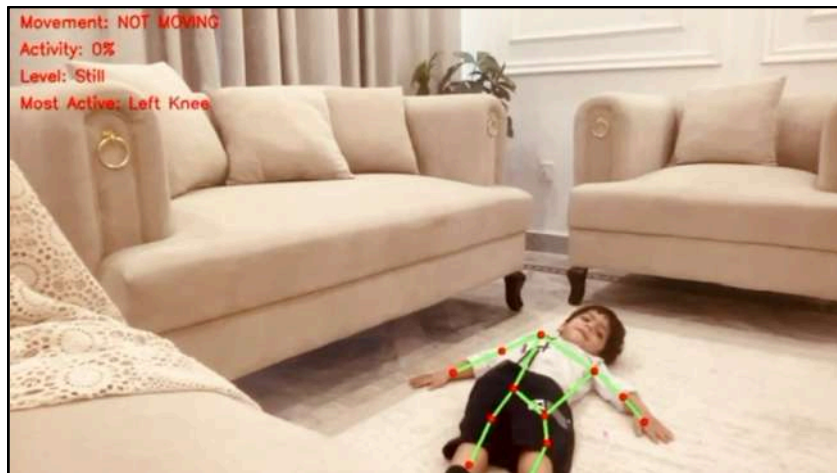


Figure 4.5: Input Image of Child in Static State

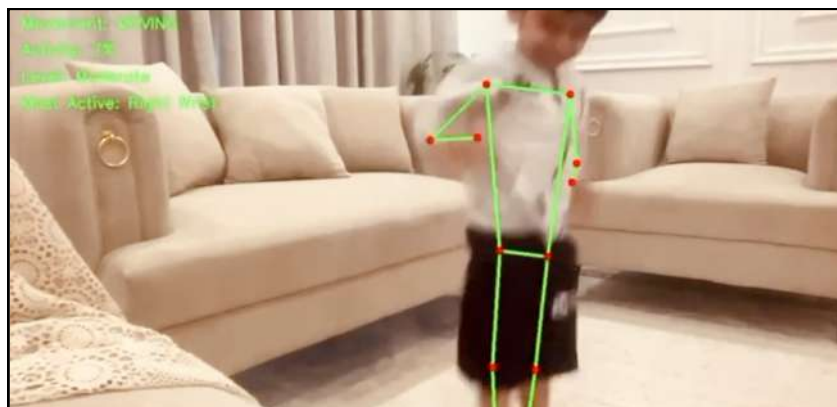


Figure 4.6: Child in Resting State

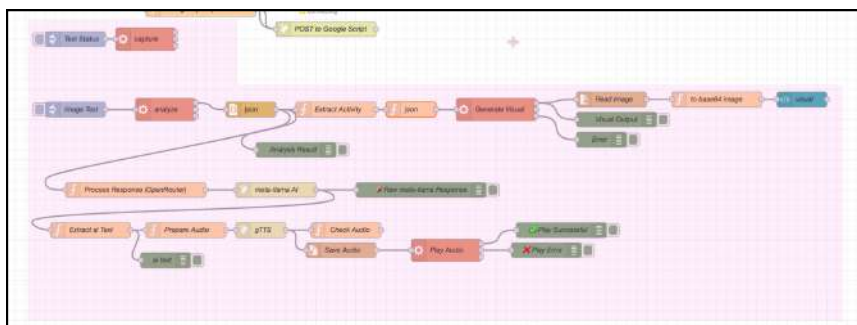


Figure 4.7: Node-RED Motivational System Workflow Overview

## 4.2.6 Node-RED Dashboard Communication Test

Real-time data show latency and configuration command reliability testing of the therapist dashboard was performed. It was shown that the sensor data displayed on the dashboard was updated at an average rate of 4.7 Hz, much higher than the 2 Hz minimum requirement. The dashboard issued configuration commands (phase transitions, PBWS level changes, session start/stop) were picked up and executed by ESP32 within 350 ms in all 50 test trials. Also not even one failed transmission was observed during the test period. Automatic data logging to Google Sheets was confirmed for all 20 simulated sessions, with complete data integrity.

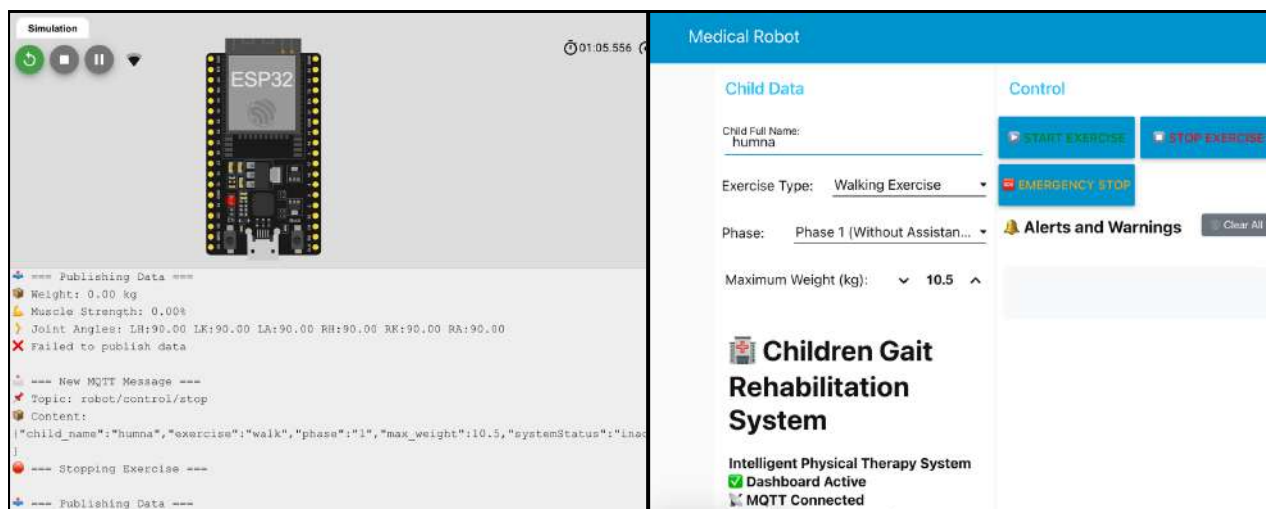


Figure 4.8: Robot Power ON/OFF Simulation

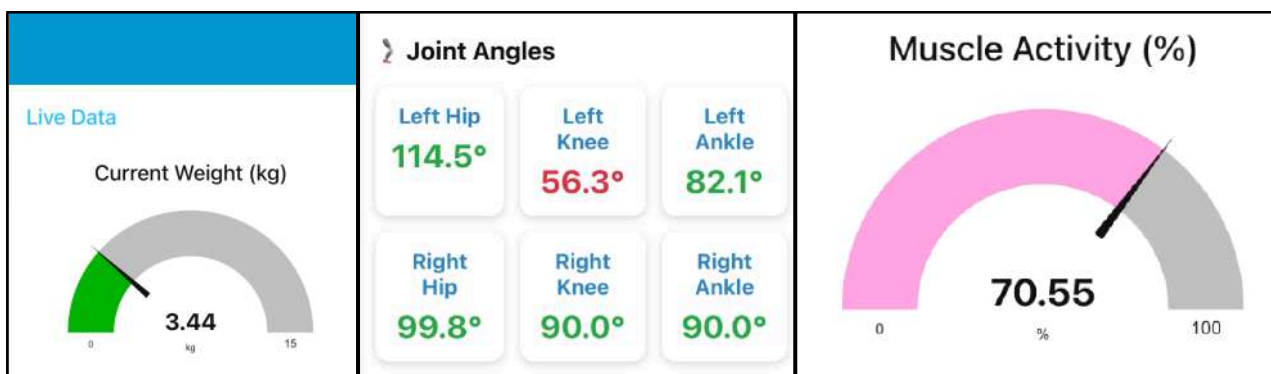


Figure 4.9: Real-Time Sensor Monitoring on Dashboard

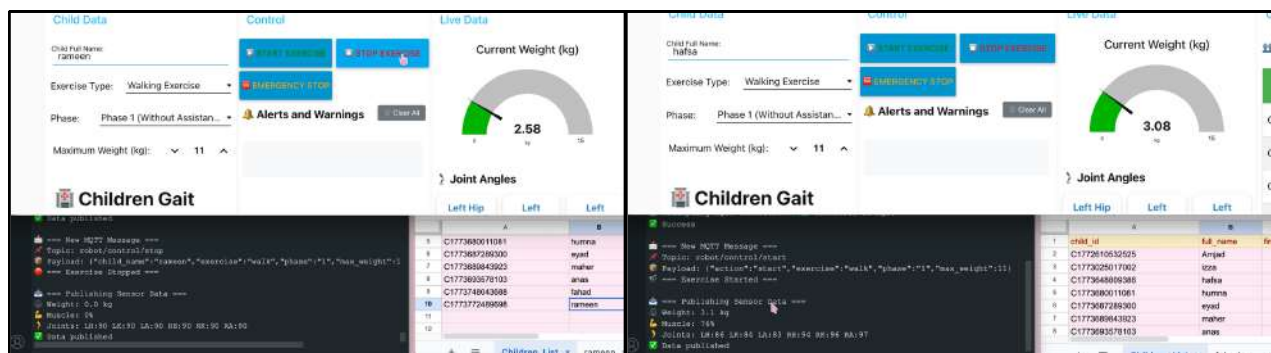


Figure 4.9: Dashboard Communication Validation

### 4.3 Integrated System Test

A complete integrated system test was conducted simulating a full 20-minute therapy session using the Group 1 protocol, with an operator manually actuating the mannequin limbs to simulate voluntary effort during Phase 2. The system successfully transitioned from Phase 1 to Phase 2 on therapist command, the assist-as-needed algorithm responded appropriately to all simulated intent signals, the motivational system delivered audio and visual feedback at appropriate intervals throughout the session, and the complete session data was logged to Google Sheets without interruption. No safety-critical failures or unexpected system shutdowns were observed.

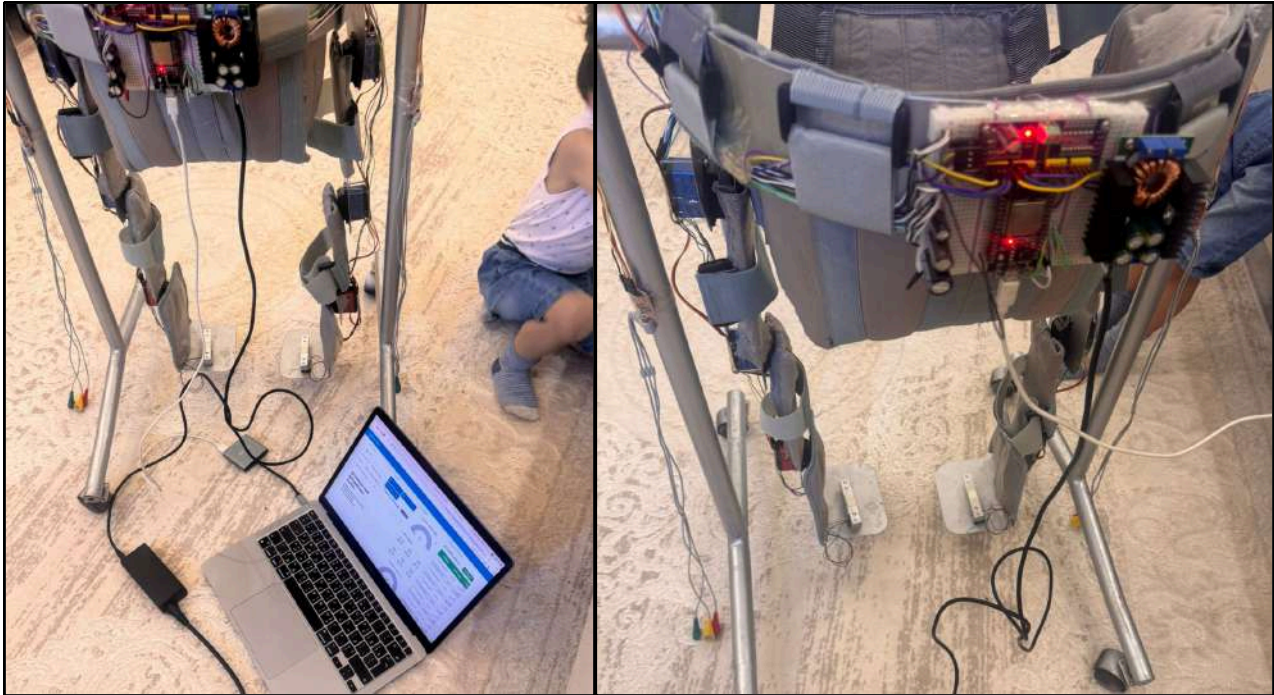


Figure 4.11: Complete System Testing Setup



Figure 4.12: Child Wearing the Exoskeleton (Lower Body View)

## **Chapter 5**

### **Conclusion and Future Recommendations**

This chapter summarizes the outcomes of the project, reflects on the degree to which the stated objectives were achieved, identifies the limitations of the current system, and proposes directions for future development.

#### **5.1 Conclusion**

This project has successfully designed, implemented, and validated a Wearable Robotic Trainer with AI-based Adaptive Motivation for Pediatric Gait Rehabilitation. The system integrates a manually fabricated stainless steel lower-limb exoskeleton, built with the help of a local metalsmith, with a comprehensive embedded sensing and control architecture based on the ESP32 microcontroller, and a high-level AI-driven motivational and monitoring system implemented entirely within Node-RED.

All five main objectives of the project have been accomplished. The exoskeleton was fabricated and appropriately sized for a three-year-old child. The dual-mode control system was implemented and tested, resulting in accurate position control in Phase 1 and reliable EMG-guided assist-as-needed behavior in Phase 2. The combined sensing system could detect movement intent and weight-bearing accurately and quickly enough to meet clinical requirements. The AI motivational pipeline, developed inside Node-RED and employing MediaPipe for movement detection, Llama for personalized text generation, and gTTS for audio synthesis, provided contextually appropriate and engaging feedback during all test sessions. The Node-RED therapist dashboard allowed for live session control and dependable data logging. The entire system made a real difference in the field of pediatric rehabilitation engineering by solving the problem of three dimensions that were previously left out: early intervention of children of approximately three years of age; structured protocol-driven therapy progression based on the CME framework; and AI-driven adaptive motivation to sustain child engagement throughout the treatment. Testing results confirm that the integrated system operates stably in a simulated therapy environment and complies with all the functional requirements set out. More generally, this work reveals that low-cost, user-specific wearable robotic trainers with smart motivational systems can be conceived and constructed within the framework of a Final Year Design Project. This indicates a realistic route towards making pediatric rehabilitation technology available in community clinics typically found in Pakistan and other developing countries.

## 5.2 Future Recommendations

We recommend several directions to extend and enhance this work:

- **Clinical Trials with Pediatric Patients:** The immediate follow-up would be to perform clinical trials with pediatric patients in a formal setting and with ethical permission, working closely with pediatric physiotherapy services. This will provide effectiveness evidence in real situations and allow adjustment of the control algorithms and motivational content according to the patient reactions for better results.
- **Machine Learning-Based Gait Intent Detection:** The present EMG threshold-based detection method can be enhanced by a well-trained machine learning model which will not only detect with better accuracy but also minimize false alarms and help in recognizing various gait phases in different patients and sessions.
- **Lightweight Frame Materials:** Changing from PLA that is 3D printed to structures made of carbon fibre reinforced polymer (CFRP) or aluminum alloy would decrease the exoskeleton weight, provide better structural rigidity, and make the device more durable.
- **Wireless EMG and Encoder Modules:** It is possible to change the existing wired sensor connections to wireless ones that have very low latency (e.g., Bluetooth Low Energy), this, besides making the device more wearable, will also eliminate the danger of the cables being tangled during the patient's active therapy sessions.
- **Expanded Age and Size Range:** Building a set of reusable structural parts will help the system support children of different age groups and with different body sizes, which means the clinical population who can benefit from the device will be considerably enlarged.
- **Integration with Electronic Health Records:** The data recording mechanism can be improved so that it can work directly with the Electronic Health Record (EHR) systems of hospitals making it easy to incorporate therapy progress data into the patient's overall clinical record, thereby helping the clinical team in monitoring long-term outcomes.
- **Gamification of the Motivational System:** Improving the motivational system by adding gaming features like virtual reward systems, story progression, and visually engaging puzzles, will definitely encourage kids more, especially in long-term therapy programs.

## References

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## Annexure A

### Complex Engineering Activities (CEA Attributes)

| Code | CEA Attribute           | Project Relevance                                                                                                                                                                                                                                                                                                  |
|------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EA1  | Range of Resources      | This project involves the management of diverse resources including financial resources (component budget of Rs 67,230), technical resources (sensors, motors, 3D printing), informational resources (research literature), and human resources (three-member project team plus supervisors).                      |
| EA2  | Level of Interaction    | The project required balancing conflicting design requirements, including component cost versus torque performance, device weight versus structural rigidity, and real-time control latency versus computational complexity, while incorporating continuous feedback from supervisors and clinical considerations. |
| EA3  | Innovation              | The project delivers a novel integration of exoskeleton hardware, EMG-guided assist-as-needed control, and an AI- based adaptive motivational system within a single device targeted at early pediatric intervention, addressing gaps not met by any existing commercial or research system.                       |
| EA4  | Consequences to Society | The project has significant positive societal implications by improving the accessibility and effectiveness of pediatric gait rehabilitation, with direct alignment with UN SDG 3 (Good Health and Well-Being), SDG 8 (Decent Work and Economic Growth), and SDG 9 (Industry, Innovation, and Infrastructure).     |
| EA5  | Familiarity             | The project required application of knowledge from multiple engineering disciplines, including embedded systems, control theory, signal processing, mechanical design, AI and automation, and human-computer interaction, in an unfamiliar and specialized domain: wearable robotics for pediatric rehabilitation. |